Asymptotic center–manifold for systems with parameter perturbations

Prabal S. Negi

Okinawa Institute of Science and Technology Graduate University, Onna, Okinawa 904-0495, Japan.

Corresponding author(s). E-mail(s): prabal.negi@oist.jp;

Abstract

Asymptotic center-manifold approximations are constructed for systems that have been perturbed slightly away from the bifurcation point in the parameter space. This is done by extending the existing system to include parameter perturbations as dynamic variables, leading to an extended critical subspace and dimensionality of the center-manifold. The extension is done in a very general setting and it is shown that the consequences of such a system extension are non-trivial. The extended critical subspace has a certain structure associated with it, which may be deduced from the original system. The application of the center-manifold theorem to the extended system leads to a graph equation which rigorously valid in the vicinity of the bifurcation point (even in parameter space). The graph equation is approximated as an asymptotic power series in the amplitudes of the critical eigenmodes leading to general expressions for restricted resolvent solutions for the various terms in the power series at each order. A Ginzburg-Landau model near bifurcation is used to demonstrate the validity of the asymptotic center-manifold in the extended space.

1 Problem Definition

1.1 The Standard System

Consider a system with its state variable denoted by $\boldsymbol{U}$, consisting of l parameters given by the vector $\boldsymbol{\mu}$, and an evolution law given by

$$\frac{\partial \boldsymbol{U}}{\partial t} = \boldsymbol{f}(\boldsymbol{U}, \boldsymbol{\mu}). \tag{1}$$

The system state $\boldsymbol{U}$ is N-dimensional where, N is assumed to be large. The evolution law may arise from a finite set of ordinary differential equations or, may be a finite representation of an infinite dimensional partial differential equation as often arises from the numerical discretization of the space-time evolution equations of physical systems. Typically, physical systems will have real parameters $\boldsymbol{\mu}$ however, allowing for complex parameters presents no additional difficulty and we assume that $\boldsymbol{\mu} \in \mathbb{C}^l$. For a certain set of parameter values $\boldsymbol{\mu} = \boldsymbol{\mu^b}$, the system is assumed to have a fixed point for a state vector $\boldsymbol{U^b}$ satisfying

$$0 = \boldsymbol{f}(\boldsymbol{U^b}, \boldsymbol{\mu^b}). \tag{2}$$

The system may be rewritten in terms of the deviation from the fixed point solution $\boldsymbol{U} = \boldsymbol{U^b} + \boldsymbol{u}$ such that

$$\begin{aligned}\frac{\partial \boldsymbol{u}}{\partial t} =& \mathbf{L}\boldsymbol{u} + \\ & [\boldsymbol{f}(\boldsymbol{U^b} + \boldsymbol{u}, \boldsymbol{\mu^b}) - \mathbf{L}\boldsymbol{u}],\end{aligned} \tag{3}$$

where, $\mathbf{L}$ is the linearized operator at the fixed point, acting on $\boldsymbol{u}$, defined by

$$\mathbf{L}\boldsymbol{u} = \left(\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{U}}\bigg|_{(\boldsymbol{U^b}, \boldsymbol{\mu^b})} \right) \cdot \boldsymbol{u}. \tag{4}$$

The right hand side in equation (3) has been divided into the linear and non-linear contributions with $\mathbf{L}(\boldsymbol{u})$ representing the linear part and the remaining terms in the square brackets representing the non-linear contributions.

Furthermore, we assume that $\boldsymbol{\mu^b}$ represents the parameters of the system at a bifurcation point. It is generically assumed that the system undergoes a multiple co-dimension bufurcation at $\boldsymbol{\mu^b}$ and that the linearized operator $\mathbf{L}$ has a discrete set of n critical eigenvalues λ^C, with zero growth rate ($\mathfrak{Re}(\lambda^C) = 0$). This includes both purely real critical eigenvalues with $\mathfrak{Re}(\lambda^C) = 0$, $\mathfrak{Im}(\lambda^C) = 0$, and purely imaginary critical eigenvalues with $\mathfrak{Re}(\lambda^C) = 0$, $\mathfrak{Im}(\lambda^C) \neq 0$. For real systems complex eigenvalues occur in conjugate pairs. Accordingly, the system has n critical eigenpairs $(\lambda_i^C, \boldsymbol{v}_i^C), i \in 1 \dots n$, where $\boldsymbol{v}_i^C$ represents the eigenvector corresponding to λ_i^C.

The adjoint of the direct linearized operator $\mathbf{L}$ is denoted as $\mathbf{L}^\dagger$ which also has n critical eigenpairs $(\lambda_i^{C*}, \boldsymbol{w}_i^C)$. The scaling of the eigenvectors of the direct and adjoint operators may be defined such that they satisfy the birthogonality condition

$$\langle \boldsymbol{w}_i^C, \boldsymbol{v}_j^C \rangle = \boldsymbol{w}_i^{C\dagger} \boldsymbol{v}_j^C = \delta_{i,j}, \tag{5}$$

where $\langle \cdot, \cdot \rangle$ is the standard inner-product and $\delta_{i,j}$ is the Kronecker-delta fuction. The superscript † is used here to indicate a complex-conjugated transpose of vectors and matrix operators. For scalars complex-conjugation is indicated by a * superscript. The superscript ^{H} which is often used for the same purpose is reserved for objects referring to harmonic forcings. One may define a matrix $\mathbf{V}$, whose columns comprise of the n critical eigenvectors of $\mathbf{L}$ and the matrix spans the invariant center subspace of the linear operator $\mathbf{L}$. The corresponding matrix for the adjoint eigenvectors is denoted $\mathbf{W}$. Due to the chosen scaling denoted by equation (5) the matrices satisfy,

$$\mathbf{W}^\dagger \mathbf{V} = \mathbf{I}_n, \tag{6}$$

where, $\mathbf{I}_n$ represents an identity matrix of order $n \times n$. The projection operator into the critical subspace (also the center subspace) is defined by $\mathbf{P} = \mathbf{V}\mathbf{W}^\dagger$, while the projection operator into its complement (stable subspace of $\mathbf{L}$) is defined by, $\mathbf{Q} = \mathbf{I} - \mathbf{V}\mathbf{W}^\dagger$.

Additionally, $\boldsymbol{\Lambda}_C$ is defined as an $n \times n$ diagonal matrix consisting of the critical eigenvalues of $\mathbf{L}$, with the implicit assumption of diagonalizability of $\mathbf{L}$ within this subspace. The ordering of the eigenvalues in $\boldsymbol{\Lambda}_C$ and eigenvectors in $\mathbf{V}$ is consistent with each other such that $\mathbf{L}\mathbf{V} = \mathbf{V}\boldsymbol{\Lambda}_C$.

1.2 The Extended System

The same system is now considered for the case of parameter perturbation so that the evolution dynamics are now evaluated at parameter values of $\boldsymbol{\mu} = \boldsymbol{\mu^b} + \boldsymbol{\nu}$, with $\boldsymbol{\nu}$ being the vector of parameter perturbations, $\boldsymbol{\nu} = [\nu_1, \nu_2, \dots, \nu_l]$. In addition, the system is forced with small-amplitude harmonic forcing of the type

$$\boldsymbol{f}_H(t) = \sum_{i=1}^{h} \boldsymbol{f}_i \theta_i(t); \qquad \theta_i = \epsilon_i e^{\lambda_i^H t}, \tag{7}$$

where, the various θ_i are treated as the dynamic variable, λ_i^H is purely imaginary, with the imaginary part representing the angular frequency of the forcing. $\boldsymbol{f}_i$ represents the shape of the i^{th} harmonic forcing and ϵ_i is a scale factor. Constant forcing terms are readily incorporated when a particular (or multiple) $\lambda_i^H = 0$. The vector of the various θ_i is denoted as $\boldsymbol{\theta} = [\theta_1; \theta_2; \dots; \theta_h]$, and the vector of angular frequencies is denoted as $\boldsymbol{\lambda}^H = [\lambda_1^H; \lambda_2^H; \dots; \lambda_h^H]$. The resulting evolution equation is

$$\frac{\partial \boldsymbol{u}}{\partial t} = \boldsymbol{f}(\boldsymbol{U^b} + \boldsymbol{u}, \boldsymbol{\mu^b} + \boldsymbol{\nu}) + \sum_{i=1}^{h} \boldsymbol{f}_i \theta_i; .$$

The diagonal matrix for the harmonic angular frequencies is represented by $\boldsymbol{\Lambda}_H = diagm(\boldsymbol{\lambda}^H)$. Similarly, $\boldsymbol{\lambda}^P = [\lambda_1^P; \lambda_2^P; \dots; \lambda_l^P]$ is defined as a vector of length l, with all $\lambda_i^P = 0$ identically, for $i \in 1 \dots l$. These may be thought of parameter eigenvalues of parameter angular frequencies depending on the point of view. In either case, these vanish identically. $\boldsymbol{\Lambda}_P = diagm(\boldsymbol{\lambda}^P)$ is defined as an $l \times l$ null matrix, to represent the trivial linear evolution matrix for the parameter variables $\boldsymbol{\nu}$. The extended system including the evolution equations for the newly introduced variables $\boldsymbol{\nu}$ and $\boldsymbol{\theta}$, henceforth referred to as the *extended variables*, is given by,

$$\frac{\partial \boldsymbol{u}}{\partial t} = \mathbf{L}\boldsymbol{u} + \mathbf{L}_{\boldsymbol{\mu}}\boldsymbol{\nu} + \mathbf{L}_{\boldsymbol{\theta}}\boldsymbol{\theta} + \boldsymbol{f}_{NL}(\boldsymbol{u},\boldsymbol{\nu}) \tag{8a}$$

$$\frac{d\boldsymbol{\nu}}{dt} = \boldsymbol{\Lambda}_P\boldsymbol{\nu} \tag{8b}$$

$$\frac{d\boldsymbol{\theta}}{dt} = \boldsymbol{\Lambda}_H\boldsymbol{\theta}, \tag{8c}$$

Here, as done earlier, the evolution equations are split into the linear and non-linear contributions. $\mathbf{L}$ is the same linearized operator as defined earlier in equation 4. $\mathbf{L}_{\boldsymbol{\mu}}$ is the linearized operator acting on the newly introduced extended variable $\boldsymbol{\nu}$, defined as

$$\mathbf{L}_{\boldsymbol{\mu}}\boldsymbol{\nu} = \left(\left. \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{\mu}} \right|_{(\boldsymbol{U^b},\boldsymbol{\mu^b})} \right) \cdot \boldsymbol{\nu}. \tag{9}$$

Similarly, $\mathbf{L}_{\boldsymbol{\theta}}$ is the linearized operator acting on $\boldsymbol{\theta}$, defined as,

$$\mathbf{L}_{\boldsymbol{\theta}}\boldsymbol{\theta} = \left(\frac{\partial \boldsymbol{f}_H}{\partial \boldsymbol{\theta}} \right) \cdot \boldsymbol{\theta}. \tag{10}$$

This is the same expression as in equation (7), just written in a manner consistent with the other terms. Finally, $\boldsymbol{f}_{NL}(\boldsymbol{u},\boldsymbol{\nu})$ represents the non-linear terms of the modified system given by

$$\begin{aligned} \boldsymbol{f}_{NL}(\boldsymbol{u},\boldsymbol{\nu}) =& \boldsymbol{f}(\boldsymbol{U^b}+\boldsymbol{u},\boldsymbol{\mu^b}+\boldsymbol{\nu}) \\ &- \mathbf{L}\boldsymbol{u} - \mathbf{L}_{\boldsymbol{\mu}}\boldsymbol{\nu}, \end{aligned} \tag{11}$$

where, the dependence on $\boldsymbol{U^b}$ and $\boldsymbol{\mu^b}$ has been suppressed for notational convenience.

An overhead hat notation, $\widehat{\cdot}$, is used to refer to variables and operators associated with the extend space, so that the new state vector, non-linear term and the linearized direct and adjoint operators are denoted by $\widehat{\boldsymbol{u}}$, $\widehat{\boldsymbol{f}}_{NL}$, $\widehat{\mathcal{L}}$ and $\widehat{\mathcal{L}^{\dagger}}$, that are defined as,

$$\widehat{\boldsymbol{u}} = \left[\boldsymbol{u};\ \boldsymbol{\nu};\ \boldsymbol{\theta}\right],$$
$$\widehat{\boldsymbol{f}}_{NL} = \left[\boldsymbol{f}_{NL};\ \mathbf{0};\ \mathbf{0}\right],$$

$$\widehat{\mathbf{L}} = \begin{bmatrix} \mathbf{L} & \mathbf{L}_{\boldsymbol{\mu}} & \mathbf{L}_{\boldsymbol{\theta}} \\ \mathbf{0} & \boldsymbol{\Lambda}_P & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{\Lambda}_H \end{bmatrix}, \quad \widehat{\mathbf{L}}^{\dagger} = \begin{bmatrix} \mathbf{L}^{\dagger} & \mathbf{0} & \mathbf{0} \\ \mathbf{L}_{\boldsymbol{\mu}}^{\dagger} & \boldsymbol{\Lambda}_P^{\dagger} & \mathbf{0} \\ \mathbf{L}_{\boldsymbol{\theta}}^{\dagger} & \mathbf{0} & \boldsymbol{\Lambda}_H^{\dagger} \end{bmatrix}, \tag{12}$$

where, the superscript ^{T} refers to vector transpose. The full system may then be represented compactly as

$$\frac{\partial \widehat{\boldsymbol{u}}}{\partial t} = \widehat{\mathbf{L}}\widehat{\boldsymbol{u}} + \widehat{\boldsymbol{f}}_{NL}. \tag{13}$$

The system defined by equation (8), or its compact form equation (13), with additional dynamical equations for the extended variables will be referred to as the *extended system*.

1.3 Extended Subspace Structure

The extended system has a larger critical subspace of dimension $m = n + l + h$, as compared to the original system due to the $l + h$ additional variables introduced. While the extension of the system seems almost trivial, the eigenvector structure in the additional critical dimensions however is far from trivial. Nevertheless, the extended eigenspace structure can be deduced directly from the properties of the original system. It turns out to be incredibly useful to analyze the structure of the extended subspace since the the adjoint extended critical subspace does have a special structure which allows the asymptotic terms of the center-manifold to be evaluated using the original linear system $\mathbf{L}$.

There are however some subtleties involved when the system extension introduces new eigenvalues that are resonant with the original linear system $\mathbf{L}$. In such a scenario the system extension leads to a generalized eigenspace for the resonant eigenvalues. The subspace structure is now evaluated in detail for both the resonant and non-resonant cases. This is first done for the extended operator $\widehat{\mathbf{L}}$ and then for the adjoint operator $\widehat{\mathbf{L}}^{\dagger}$.

1.3.1 Subspace structure for $\widehat{\mathbf{L}}$

Observe that if $(\lambda_i^C, \boldsymbol{v}_i^C)$ is a critical eigenpair of the original linearized operator $\mathbf{L}$, then its extension to $\widehat{\mathbf{L}}$ satisfies,

$$(\lambda_i^C\mathbf{I} - \widehat{\mathbf{L}})\widehat{\boldsymbol{v}}_i^C = \mathbf{0}, \tag{14a}$$

$$\widehat{\boldsymbol{v}}_i^C = \left[\boldsymbol{v}_i^C;\ \mathbf{0};\ \mathbf{0}\right], \tag{14b}$$

$\forall i \in 1,\ldots,n$, which may be verified via simple substitution. Hence all critical eigenmodes of $\mathbf{L}$ can be trivially extended to $\widehat{\mathbf{L}}$. This definition for the extension of the eigenvectors in fact applies even to modes that are not critical eigenmodes.

The additional critical eigenpairs introduced due to system extension need to be deduced. Two sets of canonical unit vectors are defined such that,

$$\boldsymbol{e}_i^P = [\overbrace{0;0;\ldots;0}^{i-1};1;\overbrace{0;0;\ldots;0}^{l-i}], \tag{15a}$$
$$\boldsymbol{e}_i^H = [\underbrace{0;0;\ldots;0}_{i-1};1;\underbrace{0;0;\ldots;0}_{h-i}]. \tag{15b}$$

If the original linear system $\mathbf{L}$ is non-singular then, the l eigenpairs generated by the system extension due to $\boldsymbol{\nu}$ have the structure of the type $(\lambda_i^P, [\boldsymbol{v}_i^P;\boldsymbol{e}_i^P;\mathbf{0}])$, which satisfy

$$(\lambda_i^P\mathbf{I}-\mathbf{L})\boldsymbol{v}_i^P = \mathbf{L}_{\boldsymbol{\mu}}(\boldsymbol{e}_i^P), \qquad (\lambda_i^P = 0) \tag{16a}$$
$$\widehat{\boldsymbol{v}}_i^P = \left[\boldsymbol{v}_i^p;\ \boldsymbol{e}_i^P;\ \mathbf{0}\right], \quad \forall i \in 1,\ldots l. \tag{16b}$$

This can be verified that $(\lambda_i^P - \widehat{\mathbf{L}})\widehat{\boldsymbol{v}}_i^P = 0$.

However, if $\mathbf{L}$ is singular then equation (16) is indeterminate and the definition needs to be modified. For the i^{th} *parameter mode*, define a vector $\boldsymbol{\gamma}_P^i \in \mathbb{C}^n$, where the j^{th} element of the vector is given by,

$$(\gamma_i^P)_j = \begin{cases} \boldsymbol{w}_j^{C\dagger}\mathbf{L}_{\boldsymbol{\mu}}(\boldsymbol{e}_i^P), & \textit{if } \lambda_i^P = \lambda_j^C \\ 0, & \textit{otherwise.} \end{cases} \tag{17}$$

$$(\lambda_i^P\mathbf{I}-\mathbf{L})\boldsymbol{v}_i^P = \mathbf{L}_{\boldsymbol{\mu}}(\boldsymbol{e}_i^P) - \mathbf{V}\gamma_i^P, \quad (\lambda_i^P = 0) \tag{18a}$$
$$\boldsymbol{w}_j^{C\dagger}\boldsymbol{v}_i^P = 0, \qquad \textit{if } \lambda_i^P = \lambda_j^C, \tag{18b}$$
$$\widehat{\boldsymbol{v}}_i^P = \left[\boldsymbol{v}_i^P;\ \boldsymbol{e}_i^P;\ \mathbf{0}\right], \qquad \forall i \in 1,\ldots l. \tag{18c}$$

In the singular case, $\widehat{\boldsymbol{v}}_i^P$ is no longer an eigenvector of $\widehat{\mathbf{L}}$, however, it is a generalized eigenvector satisfying $(\lambda_i^P\mathbf{I} - \widehat{\mathbf{L}})^2\widehat{\boldsymbol{v}}_i^P = \mathbf{0}$. Note that γ_i^P is identically zero for the non-singular case and equation (18) reduces to equation (16). Henceforth these modes shall be referred to as *parameter modes* and will be defined by equation (18), since it is the more general definition. The matrix of all the vectors $\boldsymbol{\gamma}_i^P$ is denoted $\boldsymbol{\Gamma}_P$.

Similarly, when the harmonic forcing is non-resonant, the h eigenpairs generated by system extension due to $\boldsymbol{\theta}$ have the following structure,

$$(\lambda_i^H\mathbf{I}-\mathbf{L})\boldsymbol{v}_i^H = \mathbf{L}_{\boldsymbol{\theta}}(\boldsymbol{e}_i^H), \tag{19a}$$
$$\widehat{\boldsymbol{v}}_i^H = \left[\boldsymbol{v}_i^H;\ \mathbf{0};\ \boldsymbol{e}_i^H\right], \quad \forall i \in 1,\ldots h. \tag{19b}$$

On the other hand, when the harmonic forcing is resonant with some of the angular frequencies of $\mathbf{L}$ then, analogous to the case of parameter modes, one defines,

$$(\gamma_i^H)_j = \begin{cases} \boldsymbol{w}_j^{C\dagger}\mathbf{L}_{\boldsymbol{\theta}}(\boldsymbol{e}_i^H), & \textit{if } \lambda_i^H = \lambda_j^C \\ 0, & \textit{otherwise,} \end{cases} \tag{20}$$

where, $(\gamma_i^H)_j$ is the j^{th} component of the vector $\boldsymbol{\gamma}_i^H$. The extended modes are then given by

$$(\lambda_i^H\mathbf{I}-\mathbf{L})\boldsymbol{v}_i^H = \mathbf{L}_{\boldsymbol{\theta}}(\boldsymbol{e}_i^H) - \mathbf{V}\boldsymbol{\gamma}_i^H, \tag{21a}$$
$$\boldsymbol{w}_j^{C\dagger}\boldsymbol{v}_i^H = 0, \qquad \textit{if } \lambda_i^H = \lambda_j^C, \tag{21b}$$
$$\widehat{\boldsymbol{v}}_i^H = \left[\boldsymbol{v}_i^H;\ \mathbf{0};\ \boldsymbol{e}_i^H\right], \qquad \forall i \in 1,\ldots h. \tag{21c}$$

In the resonant case, this satisfies the generalized eigenvalue equation $(\lambda_i^H\mathbf{I} - \widehat{\mathbf{L}})^2\widehat{\boldsymbol{v}}_i^H = \mathbf{0}$. Again, equation (21) reduces to equation (19) when the forcing is non-resonant. Henceforth these modes will be referred to as *forcing modes* and are defined by the more general definition given by equation (21). The matrix of all the vectors $\boldsymbol{\gamma}_i^H$ is denoted $\boldsymbol{\Gamma}_H$. The critical eigenspace of $\widehat{\mathbf{L}}$ can be represented by three matrices

$$\widehat{\mathbf{V}}_C = \begin{bmatrix}\mathbf{V}\\ \mathbf{0}\\ \mathbf{0}\end{bmatrix}, \quad \widehat{\mathbf{V}}_P = \begin{bmatrix}\mathbf{V}_P\\ \mathbf{I}\\ \mathbf{0}\end{bmatrix}, \quad \widehat{\mathbf{V}}_H = \begin{bmatrix}\mathbf{V}_H\\ \mathbf{0}\\ \mathbf{I}\end{bmatrix}, \tag{22}$$

where, $\mathbf{V}$ is the matrix of the critical eigenvectors of $\mathbf{L}$. The matrix $\mathbf{V}_P$ comprises of the l vectors $\boldsymbol{v}_i^P$ of the parameter modes, and, $\mathbf{V}_H$ comprises of the h vectors $\boldsymbol{v}_i^H$ of the forcing modes. The matrix of vectors of the extended eigenspace is denoted by, $\widehat{\mathbf{V}} = [\widehat{\mathbf{V}}_C, \widehat{\mathbf{V}}_P, \widehat{\mathbf{V}}_H]$.

1.3.2 Subspace structure for $\widehat{\mathbf{L}}^\dagger$

The eigenspace structure for the adjoint problem is important to consider since these vectors define the projection operators into the extended critical subspace. The critical eigenmodes of the $\widehat{\mathbf{L}}^\dagger$ can be extended via the following definition,

$$(\lambda_i^{C*}\mathbf{I} - \mathbf{L}^\dagger)\boldsymbol{w}_i^C = 0, \qquad \forall i \in 1, \ldots n, \tag{23a}$$
$$\widehat{\boldsymbol{w}}_i^C = \left[\boldsymbol{w}_i^C;\ \boldsymbol{\zeta}_i^P;\ \boldsymbol{\zeta}_i^H\right], \tag{23b}$$

with vectors $\boldsymbol{\zeta}_i^P$ and $\boldsymbol{\zeta}_i^H$ defined such that the j^{th} component of $\boldsymbol{\zeta}_i^P$ given by,

$$(\boldsymbol{\zeta}_i^P)_j = \begin{cases} (\mathbf{L}_{\boldsymbol{\mu}}^\dagger \boldsymbol{w}_i^C)_j/(\lambda_i^{C*} - \lambda_j^{P*}); & \textit{if } \lambda_i^{C*} \neq \lambda_j^{P*}, \\ 0; & \textit{if } \lambda_i^{C*} = \lambda_j^{P*}, \end{cases} \tag{24}$$

$\forall j \in 1, \ldots, l$, and, the j^{th} component of $\boldsymbol{\zeta}_i^H$ is given by,

$$(\boldsymbol{\zeta}_i^H)_j = \begin{cases} (\mathbf{L}_{\boldsymbol{\theta}}^\dagger \boldsymbol{w}_i^C)_j/(\lambda_i^{C*} - \lambda_j^{H*}); & \textit{if } \lambda_i^{C*} \neq \lambda_j^{H*}, \\ 0; & \textit{if } \lambda_i^{C*} = \lambda_j^{H*}, \end{cases} \tag{25}$$

$\forall j \in 1, \ldots, h$. Here, $(\mathbf{L}_{\boldsymbol{\mu}}^\dagger \boldsymbol{w}_i^C)_j$ refers to the j^{th} component of the vector $(\mathbf{L}_{\boldsymbol{\mu}}^\dagger \boldsymbol{w}_i^C)$, and likewise for $(\mathbf{L}_{\boldsymbol{\theta}}^\dagger \boldsymbol{w}_i^C)_j$. It may be verified that the definition is consistent with the eigenvalue equation, $(\lambda_i^{C*}\mathbf{I} - \widehat{\mathbf{L}}^\dagger)\widehat{\boldsymbol{w}}_i^C = 0$, for the non-resonant cases, and satisfies generalized eigenvalue equation, $(\lambda_i^{C*}\mathbf{I} - \widehat{\mathbf{L}}^\dagger)^2\widehat{\boldsymbol{w}}_i^C = 0$, for the resonant cases.

The adjoint parameter modes are given simply by,

$$\widehat{\boldsymbol{w}}_i^P = \left[\mathbf{0};\ \boldsymbol{e}_i^P;\ \mathbf{0}\right], \qquad \forall i \in 1, \ldots l, \tag{26}$$

which satisfies $(\lambda_i^{P*}\mathbf{I} - \widehat{\mathbf{L}}^\dagger)\widehat{\boldsymbol{w}}_i^P = 0$. And the adjoint forcing modes are given by,

$$\widehat{\boldsymbol{w}}_i^H = \left[\mathbf{0};\ \mathbf{0};\ \boldsymbol{e}_i^H\right], \qquad \forall i \in 1, \ldots h, \tag{27}$$

which satisfies $(\lambda_i^{H*}\mathbf{I} - \widehat{\mathbf{L}}^\dagger)\widehat{\boldsymbol{w}}_i^H = 0$.

The extended eigenspace of $\widehat{\mathbf{L}}^\dagger$ can be represented with matrices as

$$\widehat{\mathbf{W}}_C = \begin{bmatrix} \mathbf{W} \\ \mathbf{Z}_P \\ \mathbf{Z}_H \end{bmatrix}, \quad \widehat{\mathbf{W}}_P = \begin{bmatrix} \mathbf{0} \\ \mathbf{I} \\ \mathbf{0} \end{bmatrix}, \quad \widehat{\mathbf{W}}_H = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{I} \end{bmatrix}, \tag{28}$$

where, $\mathbf{W}$ is the matrix of the adjoint critical eigenvectors of $\mathbf{L}^\dagger$. The matrix $\mathbf{Z}_P$ comprises of the l vectors $\boldsymbol{\zeta}_i^P$ and, $\mathbf{Z}_H$ comprises of the h vectors $\boldsymbol{\zeta}_i^H$. The matrix of vectors of the critical adjoint eigenspace is denoted by, $\widehat{\mathbf{W}} = [\widehat{\mathbf{W}}_C, \widehat{\mathbf{W}}_P, \widehat{\mathbf{W}}_H]$. It can be verified that,

$$\widehat{\mathbf{W}}^\dagger \widehat{\mathbf{V}} = \mathbf{I}_m, \tag{29}$$

where, $m = n + l + h$. This is the analogue of equation (6), and is simply a statement of the biorthogonality relation between the direct and adjoint eigenvectors in the extended space. The projection matrices into the critical and stable subspaces of $\widehat{\mathbf{L}}$ are given by,

$$\widehat{\mathbf{P}} = \widehat{\mathbf{V}}\widehat{\mathbf{W}}^\dagger \tag{30}$$
$$\implies \widehat{\mathbf{P}} = \widehat{\mathbf{V}}_{\mathbf{C}}\widehat{\mathbf{W}}_C^\dagger + \widehat{\mathbf{V}}_{\mathbf{P}}\widehat{\mathbf{W}}_P^\dagger + \widehat{\mathbf{V}}_{\mathbf{H}}\widehat{\mathbf{W}}_H^\dagger,$$
$$\widehat{\mathbf{Q}} = \mathbf{I} - \widehat{\mathbf{V}}\widehat{\mathbf{W}}^\dagger, \tag{31}$$
$$\implies \widehat{\mathbf{Q}} = \mathbf{I} - \widehat{\mathbf{V}}_{\mathbf{C}}\widehat{\mathbf{W}}_C^\dagger - \widehat{\mathbf{V}}_{\mathbf{P}}\widehat{\mathbf{W}}_P^\dagger - \widehat{\mathbf{V}}_{\mathbf{H}}\widehat{\mathbf{W}}_H^\dagger.$$

Finally, we have the generalized eigenvalue relation,

$$\widehat{\mathbf{L}}\widehat{\mathbf{V}} = \widehat{\mathbf{V}}\widehat{\mathbf{K}}, \tag{32}$$

with,

$$\widehat{\mathbf{K}} = \widehat{\mathbf{W}}^\dagger \widehat{\mathbf{L}}\widehat{\mathbf{V}} = \begin{bmatrix} \mathbf{\Lambda}_C & \mathbf{\Gamma}_P & \mathbf{\Gamma}_H \\ \mathbf{0} & \mathbf{\Lambda}_P & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{\Lambda}_H \end{bmatrix}, \tag{33}$$

being the reduced matrix representing the oblique projection of the extended linear operator $\widehat{\mathbf{L}}$ onto the critical subspace. Obviously it is a complex matrix of size $m \times m$. The diagonal elements of $\widehat{\mathbf{K}}$ represent the eigenvalues of the reduced system, that are the critical eigenvalues of the extended system, and may be represented by the vector $\widehat{\boldsymbol{\lambda}} = diag(\widehat{\mathbf{K}})$.

Note that due to the special structure of the adjoint parameter and forcing modes, the projection of the extended nonlinearity $\widehat{\boldsymbol{f}}_{NL}$ onto the subspace spanned by the (direct) parameter and forcing modes vanishes identically. That is to say,

$$\widehat{\mathbf{W}}_P^\dagger \widehat{\boldsymbol{f}}_{NL} = \mathbf{0}, \qquad \widehat{\mathbf{W}}_H^\dagger \widehat{\boldsymbol{f}}_{NL} = \mathbf{0}. \tag{34}$$

This completes the extension of the system and the subspace structure for both the direct and adjoint systems can be readily identified. While the structure of the extended subspaces has been evaluated, reformulation of the problem as an extended system may have detrimental practical implications when considering large systems. Practical methods for solving such systems often rely on numerical algorithms that exploit inherent structure of the problem (Hermicity for conjugate-gradient solvers for example). In such a scenario, additional degrees of freedom can potential destroy the system symmetries and may require substantial changes in the numerical approach to the system solution. This would be particularly true of asymptotic approaches that require resolvent calculations [1, 2, 3, 4, 5, 6, 7, 8, 9], which itself is a computationally challenging task for large systems. Such a scenario would be highly undesireable. However, as it turns out, at least some of the parameterizations involved in the evaluation of center-manifolds do not in fact require the use of the extended system after the initial evaluation of the extended tangent space. All higher order calculations can be performed using the original linear system $\mathbf{L}$. This is clarified in the next section where the parameterization and its asymptotic terms are explicitly built. First however, we prove the following results which will come in handy.

Lemma 1. *Given an extended system described by equation (8), with a linearized operator $\widehat{\mathbf{L}}$, given by equation (12), if a vector, $\widehat{\boldsymbol{u}} = [\boldsymbol{u}; \boldsymbol{\nu}; \boldsymbol{\theta}]$, is such that its projection onto the center-subspace of $\widehat{\mathbf{L}}$ vanishes, then, the extended variables $\boldsymbol{\nu}$ and $\boldsymbol{\theta}$ must also vanish identically for $\widehat{\boldsymbol{u}}$.*

Proof A vector $\widehat{\boldsymbol{u}}$ whose components along the center-subspace vanish implies that $\widehat{\mathbf{W}}^\dagger \widehat{\boldsymbol{u}} = \mathbf{0}$, where $\widehat{\mathbf{W}} = [\widehat{\mathbf{W}}_C, \widehat{\mathbf{W}}_P, \widehat{\mathbf{W}}_H]$, are as deduced in sections 1.2 and 1.3.

$$\therefore \widehat{\mathbf{W}}^\dagger \widehat{\boldsymbol{u}} = \begin{bmatrix} \mathbf{W}^\dagger \boldsymbol{u} + \mathbf{Z}_P^\dagger \boldsymbol{\nu} + \mathbf{Z}_H^\dagger \boldsymbol{\theta} \\ \mathbf{I}^\dagger \boldsymbol{\nu} \\ \mathbf{I}^\dagger \boldsymbol{\theta} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}.$$

For this to hold,

$$\mathbf{I}^\dagger \boldsymbol{\nu} = \mathbf{0}, \implies \boldsymbol{\nu} = \mathbf{0},$$
$$\mathbf{I}^\dagger \boldsymbol{\theta} = \mathbf{0}, \implies \boldsymbol{\theta} = \mathbf{0}.$$

Hence, both the extended variables $\boldsymbol{\nu}$ and $\boldsymbol{\theta}$ vanish identically. □

Corollary 1. *Given an extended system described by equation (8), with a linearized operator $\widehat{\mathbf{L}}$, given by equation (12), if a vector, $\widehat{\boldsymbol{u}} = [\boldsymbol{u}; \boldsymbol{\nu}; \boldsymbol{\theta}]$, is such that its projection onto the center-subspace of $\widehat{\mathbf{L}}$ vanishes, then, the projection of $\boldsymbol{u}$ onto the center-subspace of $\mathbf{L}$ also vanishes.*

Proof For a vector $\widehat{\boldsymbol{u}}$ that satisfies $\widehat{\mathbf{W}}^\dagger \widehat{\boldsymbol{u}} = \mathbf{0}$, by lemma 1, both $\boldsymbol{\nu}$ and $\boldsymbol{\theta}$ vanish identically.

$$\therefore \widehat{\mathbf{W}}^\dagger \widehat{\boldsymbol{u}} = \begin{bmatrix} \mathbf{W}^\dagger \boldsymbol{u} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix},$$

which implies $\mathbf{W}^\dagger \boldsymbol{u} = \mathbf{0}$. However, $\mathbf{W}$ spans the center-subspace of $\mathbf{L}^\dagger$, and therefore $\mathbf{W}^\dagger \boldsymbol{u}$ is the projection of $\boldsymbol{u}$ onto the center-subspace of $\mathbf{L}$, which vanishes as required. □

A proof of corollary 1 in the case of infinite dimensional systems with parameter extensions can be found in [4].

2 Center Manifold for the extended system

In the context of invariant manifold theory, several previous approaches for the approximation of such manifolds can be effectively viewed as instances of what is referred to as the parametrization method. The framework was introduced in a series of papers by Cabré et al. [10, 11, 2], although, a particular instance of such an approach was presented earlier by Coullet and Spiegel [1] for the evaluation of the normal form of center manifolds. Nevertheless, subsequent works by Carini, Auteri, and Giannetti [5], Jain and Haller [8], Negi [9] can be fruitfully viewed through the lens of the parameterization method. Within this framework the reduced representation of the invariant manifold being approximated can be chosen in an arbitrary manner depending on the particular requirements of the reduction being sought. Amongst the plethora of arbitrary choices, the normal form is an important representation that is often sought since it represents the simplest form of the reduced equations. In the next subsection the normal form representation for the center-manifold of the extended system is deduced.

To simplify notation for multiple indices used hereafter, a multi-dimensional index set

$\mathcal{I}\{(k),[a,b]\}$ is introduced, which is an ordered set of all k-tuples $(i_1, i_2 \ldots, i_k)$, defined as,

$$\begin{aligned}\mathcal{I}\{(k),[a,b]\} \coloneqq \{&(i_1, i_2 \ldots, i_k) \mid a, b \in \mathbb{N},\\ &i_1, i_2 \ldots, i_k \in \mathbb{N},\\ &a \leq i_i \leq b,\ i_1 \leq i_2 \leq b,\\ &\ldots,\ i_{k-1} \leq i_k \leq b.\},\end{aligned} \tag{35}$$

where, $i_1, i_2, \ldots, i_k$, are the index variables and $[a,b]$ specify the lower and upper bounds of the indices. Obviously the indices i_1 *etc.* are dummy indices however their relative position is important since the higher index always starts from the previous index. The set is ordered in the sense that the first element of the $\mathcal{I}\{(k),[a,b]\}$ is always $(i_1, i_2 \ldots, i_k) = (a, a, \ldots, a)$, the last element is $(i_1, i_2 \ldots, i_k) = (b, b, \ldots, b)$ and it is the last index (i_k) which iterates the fastest. Hence the index set $\mathcal{I}\{(3),[1,3]\}$ is the set containing the tuples,

$$\begin{aligned}(i_1, i_2, i_3) \in \{&(1,1,1), (1,1,2), (1,1,3),\\ &(1,2,2), (1,2,3), (1,3,3)\\ &(2,2,2), (2,2,3), (2,3,3)\\ &(3,3,3)\} \quad .\end{aligned}$$

Similarly, multiple nested summations are denoted as,

$$\sum_{(k)}^{[a,b]} = \sum_{i_1=a}^{b} \sum_{i_2=i_1}^{b} \cdots \sum_{i_k=i_{k-1}}^{b} \tag{36}$$

As before, the higher index starts from the previous index. In principle the ordering of the tuples in a particular index set does not matter to the overall results however, keeping the ordering in mind helps to elucidate the structure of the asymptotic problem as will be shown subsequently.

For the asymptotic expansions we will typically have $a = 1$ and $b = m$, where m is the dimension of the manifold. For such cases the number of k-tuples in a multi-index tuple set $\mathcal{I}\{(k),[1,m]\}$ is analytically known to be $T_m^k = C(m+k-1, k)$, where,

$$C(p,q) = \binom{p}{q} = \frac{p!}{q!(p-q)!}, \tag{37}$$

defines the binomial coefficient of p and q.

2.1 Normal Form

One may find the exposition on the implementation of the parameterization method in previous works of Haro et al. [7] and Jain and Haller [8]. To be clear, the application of the parameterization method to an extended system like equation (8) follows the same steps as outlined in those earlier works. However, many of the essential steps are repeated here to bring out the structure of system of equations at each asymptotic order in the context of system extension.

Consider the extended system as defined by equation (8), represented in a compact form by equation (13), which has an m-dimensional critical subspace $(m = n + l + h)$. Assume $\mathbf{x} = [x_1; x_2; \ldots; x_m]$ is an m-dimensional vector of coordinates that parameterize the center-manifold of the extended system. The evolution of the system along the center-manifold in the reduced coordinates is assumed to be given by $\dot{\mathbf{x}} = \mathcal{G}(\mathbf{x})$, where, $\mathcal{G}$ is a vector valued function. The system solution on the center-manifold corresponding to a point $\mathbf{x}$ of the reduced system is given by $\mathcal{Y}(\mathbf{x})$. At this stage neither $\mathcal{G}(\mathbf{x})$ nor $\mathcal{Y}(\mathbf{x})$ have been specified in any form, since, the parameterization can be arbitrarily chosen. In order to obtain a normal form of the reduced system one may expand both $\mathcal{G}(\mathbf{x})$ and $\mathcal{Y}(\mathbf{x})$ as a formal power series in $\mathbf{x}$ and constrain the asymptotic form of $\mathcal{G}$ to be in a normal form for the parameterized coordinates $\mathbf{x}$. The asymptotic form of $\mathcal{Y}$ is obtained by introducing the asymptotic expansions into equation (13) and solving the terms order-by-order.

In what follows, Einstein summation is *not* assumed for repeated indices unless otherwise specified. We assume the power series expansions as,

$$\begin{aligned}\mathcal{G}(\mathbf{x}) = &\sum_{(1)}^{[1,m]} \mathbf{g}_{i_1} x_{i_1} + \sum_{(2)}^{[1,m]} \mathbf{g}_{i_1,i_2} x_{i_1} x_{i_2} + \\ &\sum_{(3)}^{[1,m]} \mathbf{g}_{i_1,i_2,i_3} x_{i_1} x_{i_2} x_{i_3} + \ldots,\end{aligned} \tag{38}$$

and,

$$\mathcal{Y}(\mathbf{x}) = \sum_{(1)}^{[1,m]} \widehat{\mathbf{y}}_{i_1} x_{i_1} + \sum_{(2)}^{[1,m]} \widehat{\mathbf{y}}_{i_1,i_2} x_{i_1} x_{i_2} + \sum_{(3)}^{[1,m]} \widehat{\mathbf{y}}_{i_1,i_2,i_3} x_{i_1} x_{i_2} x_{i_3} + \dots. \tag{39}$$

Here the terms $\mathbf{g}_{i_1,i_2,i_3,\dots}$ are m-dimensional vectors while, $\widehat{\mathbf{y}}_{i_1,i_2,i_3,\dots}$ are vectors of size $(N + l + h)$, consistent with the total dimension of the extended system. Given the asymptotic form of $\mathcal{Y}(\mathbf{x})$, one automatically obtains the nonlinear term $\widehat{\boldsymbol{f}}_{NL}$ as a power series in $\mathbf{x}$. This can be represented as,

$$\widehat{\boldsymbol{f}}_{NL} = \sum_{(2)}^{[1,m]} \widehat{\mathbf{f}}_{i_1,i_2} x_{i_1} x_{i_2} + \sum_{(3)}^{[1,m]} \widehat{\mathbf{f}}_{i_1,i_2,i_3} x_{i_1} x_{i_2} x_{i_3} + \dots, \tag{40}$$

which is obtained via the collection of terms with similar powers.

Substitution of the power series expansions into equation (13) results in

$$\sum_{\alpha=1}^{m} \frac{\partial}{\partial x_\alpha} \Big(\sum_{(1)}^{[1,m]} \widehat{\mathbf{y}}_{i_1} x_{i_1} + \dots \Big) \Big(\sum_{(1)}^{[1,m]} \mathbf{g}^{\alpha}_{j_1} x_{j_1} + \dots \Big) = \widehat{\mathbf{L}} \Big(\sum_{(1)}^{[1,m]} \widehat{\mathbf{y}}_{i_1} x_{i_1} + \dots \Big) + \Big(\sum_{(2)}^{[1,m]} \widehat{\boldsymbol{f}}_{i_1,i_2} x_{i_1} x_{i_2} + \dots \Big). \tag{41}$$

Here $\mathbf{g}^{\alpha}_{j_1}$ represents the α^{th} element of the vector $\mathbf{g}_{j_1}$. At order k, one obtains $m \times T^k_m$ unknowns which define the elements of the vectors $\mathbf{g}_{i_1,\dots,i_k}$, and the corresponding T^k_m unknown vectors $\widehat{\mathbf{y}}_{i_1,\dots,i_k}$. The definitions of the two sets of unknowns are interdependent since the form of the reduced representation $\mathcal{G}(\mathbf{x})$ determines the map $\mathcal{Y}(\mathbf{x})$ from the reduced coordinates to the full solution (and vice versa).

2.1.1 Asymptotic terms of $\mathcal{O}(|\boldsymbol{x}|)$

At first order the non-linear terms do not contribute and one simply obtains,

$$\sum_{(1)}^{[1,m]} \Big(\sum_{\alpha=1}^{m} \widehat{\mathbf{y}}_{\alpha} \mathbf{g}^{\alpha}_{i_1} - \widehat{\mathbf{L}} \widehat{\mathbf{y}}_{i_1} \Big) x_{i_1} = \mathbf{0}. \tag{42}$$

Here, the first restrictions on the asymptotic form of $\mathcal{G}$ are applied, associating the unknown terms $\widehat{\mathbf{y}}_{i_1}, \dots$ and $\mathbf{g}_{i_1}, \dots$ with the generalized eigenvalue problem such that,

$$\big[\widehat{\mathbf{y}}_{i_1}, \widehat{\mathbf{y}}_{i_2}, \dots, \widehat{\mathbf{y}}_{i_m}\big] = \widehat{\mathbf{V}}, \tag{43a}$$
$$\big[\mathbf{g}_{i_1}, \mathbf{g}_{i_2}, \dots, \mathbf{g}_{i_m}\big] = \widehat{\mathbf{K}}, \tag{43b}$$

reduces equation (42) to equation (32). This then satisfies the tangency condition of the invariant manifold, *i.e.*, at the origin the center-manifold is tangent to the center subspace. Subsequently, $\widehat{K}_{i,j}$ will be used for all the first order terms of the type $\mathbf{g}^{i}_{j}$ and the first order vectors $\widehat{\mathbf{y}}_i$ will be replaced by the respective eigenvectors $\widehat{\mathbf{v}}_i$. The reduced matrix $\widehat{\mathbf{K}}$ is responsible for the cross-coupling of different unknowns at each order of the asymptotic expansion.

2.1.2 Asymptotic terms of $\mathcal{O}(|\boldsymbol{x}|^2)$

At second order the non-linear terms contribute and one obtains the following relations

$$\sum_{(2)}^{[1,m]} \sum_{j=1}^{m} \Big(\widehat{K}_{i_1,j} x_{i_2} x_j + \widehat{K}_{i_2,j} x_{i_1} x_j \Big) \widehat{\mathbf{y}}_{i_1,i_2} + \sum_{(2)}^{[1,m]} \sum_{j=1}^{m} \mathbf{g}^{j}_{i_1,i_2} \widehat{\mathbf{v}}_j x_{i_1} x_{i_2} - \sum_{(2)}^{[1,m]} \widehat{\mathbf{L}} \widehat{\mathbf{y}}_{i_1,i_2} x_{i_1} x_{i_2} = \sum_{(2)}^{[1,m]} \widehat{\mathbf{f}}_{i_1,i_2} x_{i_1} x_{i_2}. \tag{44}$$

Collecting the expressions for a particular asymptotic term $x_a x_b$, where $(a, b) \in \boldsymbol{\mathcal{I}}\{(2), [1, m]\}$, equation (44) reduces to,

$$
\begin{aligned}
&\left[\sum_{i=1}^{a} \widehat{K}_{i,b}\widehat{\mathbf{y}}_{i,a} + \sum_{i=1}^{b} \widehat{K}_{i,a}\widehat{\mathbf{y}}_{i,b} - \sum_{i=1}^{a} \widehat{K}_{i,b}\widehat{\mathbf{y}}_{i,a}\delta_b^a\right] + \\
&\left[\sum_{i=a}^{m} \widehat{K}_{i,b}\widehat{\mathbf{y}}_{a,i} + \sum_{i=b}^{m} \widehat{K}_{i,a}\widehat{\mathbf{y}}_{b,i} - \sum_{i=b}^{m} \widehat{K}_{i,a}\widehat{\mathbf{y}}_{b,i}\delta_b^a\right] \\
&+ \sum_{j=1}^{m} \mathbf{g}_{a,b}^{j}\widehat{\mathbf{v}}_j - \widehat{\mathbf{L}}\widehat{\mathbf{y}}_{a,b} = \widehat{\mathbf{f}}_{a,b}.
\end{aligned}
\tag{45}
$$

At second order one obtains T_m^k equations which collectively form the system of equations for all $\widehat{\mathbf{y}}_{i_1,i_2}$ vectors that needs to be solved. Several important properties of equation (45) need to be pointed out that have implications in terms of practial application and implementations.

First, the various unknown vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ are coupled due to the off-diagonal terms in $\widehat{\mathbf{K}}$, which appear in the first two bracketted terms in equation (45). Recall that we are considering the tuples in our index set to be ordered so that the vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ can also be ordered in the same manner. For any index $(a,b) \in \boldsymbol{\mathcal{I}}\{(2),[1,m]\}$ (and assuming $b > a$) the bracketted terms in equation (45) can be rewritten as,

$$
\begin{aligned}
&\overbrace{\sum_{i=1}^{a}\left(\widehat{K}_{i,b}\widehat{\mathbf{y}}_{i,a} - \widehat{K}_{i,b}\widehat{\mathbf{y}}_{i,a}\delta_b^a\right)}^{P1} + \overbrace{\sum_{i=1}^{a-1} \widehat{K}_{i,a}\widehat{\mathbf{y}}_{i,b}}^{P2} + \\
&\overbrace{\widehat{K}_{a,a}\widehat{\mathbf{y}}_{a,b}}^{D1} + \overbrace{\sum_{i=a+1}^{b} \widehat{K}_{i,a}\widehat{\mathbf{y}}_{i,b}}^{S1} + \\
&\overbrace{\sum_{i=a}^{b-1} \widehat{K}_{i,b}\widehat{\mathbf{y}}_{a,i}}^{P3} + \overbrace{\widehat{K}_{b,b}\widehat{\mathbf{y}}_{a,b}}^{D2} + \overbrace{\sum_{i=b+1}^{m} \widehat{K}_{i,b}\widehat{\mathbf{y}}_{a,i}}^{S2} + \\
&\overbrace{\sum_{i=b}^{m}\left(\widehat{K}_{i,a}\widehat{\mathbf{y}}_{b,i} - \widehat{K}_{i,a}\widehat{\mathbf{y}}_{b,i}\delta_b^a\right)}^{S3}.
\end{aligned}
\tag{46}
$$

The terms marked as $P1, P2, P3$ couple to vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ that precede $\widehat{\mathbf{y}}_{a,b}$ in the ordering, the terms marked $D1, D2$ couple to $\widehat{\mathbf{y}}_{a,b}$, and the terms marked as $S1, S2, S3$ couple to vectors that succeed $\widehat{\mathbf{y}}_{a,b}$ in the ordering. Notice that all the

$$
\begin{pmatrix}
\widehat{K}_{1,1} & \cdots & \left[\mathbf{P}\right] & & \left[\mathbf{P}\right] & & \\
\vdots & \ddots & & & & & \\
 & \cdots & \widehat{K}_{a,a} & \cdots & & & \\
 & & \left[\mathbf{S}\right] & \ddots & & & \\
 & & & \cdots & \widehat{K}_{b,b} & \cdots & \\
 & & & & \left[\mathbf{S}\right] & \ddots & \vdots \\
 & & & & & \cdots & \widehat{K}_{m,m}
\end{pmatrix}
$$

Fig. 1: Coupling of the vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ with the $\widehat{\mathbf{K}}$ matrix for a given (a,b).

succeeding terms in $S1, S2, S3$ are always multiplied by elements of the $\widehat{\mathbf{K}}$ matrix that are below the main diagonal. However, since $\widehat{\mathbf{K}}$ has an upper triangular structure, all terms marked $S1, S2, S3$ vanish. Therefore any vector $\widehat{\mathbf{y}}_{a,b}$ only couples with vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ which precede it in the ordering. Thus the resulting system of equations for the T_m^2 unknown fields at second order is lower triangular and even though the fields are coupled with each other, they can still be solved for one at a time with back substitution. The splitting of terms in equation (46) is done for $b > a$ however, the same conclusion applies even for the case of $a = b$ and the resulting system of equations is always lower triangular.

A visual representation of this cross coupling is shown in figure 1. For a given index $(a,b) \in \boldsymbol{\mathcal{I}}\{(2),[1,m]\}$, the elements of the matrix $\widehat{\mathbf{K}}$ marked by $\mathbf{P}$ multiply vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ that precede $\widehat{\mathbf{y}}_{a,b}$ in the ordering. Similarly, the elements marked by $\mathbf{S}$ multiply the vectors $\widehat{\mathbf{y}}_{i_1,i_2}$ which succeed $\widehat{\mathbf{y}}_{a,b}$ in the ordering. The diagonal terms $\widehat{K}_{a,a}$ and $\widehat{K}_{b,b}$ always multiply $\widehat{\mathbf{y}}_{a,b}$. Obviously, when $\widehat{\mathbf{K}}$ is upper triangular, the contributions of the succeeding elements will always vanish. This lower triangular structure of the resulting system of equations for $\widehat{\mathbf{y}}_{i_1,i_2}$ is not confined to the second-order approximation but in fact persists even for higher order asymptotic approximations so that at all orders, the system of equations can always be solved sequentially, one vector at a time. This is an important aspect particularly for large systems since the computational cost of a simultaneous coupled system solve can be prohibitive (as T_m^k becomes large).

To keep the notation compact, the non-vanishing off-diagonal coupling terms will be represented as $\widehat{\mathbf{h}}_{a,b}$, defined for second order as,

$$\widehat{\mathbf{h}}_{a,b} = \begin{cases} \sum_{i=1}^{a} \left(\widehat{K}_{i,b}\widehat{\mathbf{y}}_{i,a} - \widehat{K}_{i,b}\widehat{\mathbf{y}}_{i,a}\delta_b^a \right) \\ + \sum_{i=1}^{a-1} \widehat{K}_{i,a}\widehat{\mathbf{y}}_{i,b} + \sum_{i=a}^{b-1} \widehat{K}_{i,b}\widehat{\mathbf{y}}_{a,i} \end{cases} \tag{47}$$

Due to the diagonal terms of $\widehat{\mathbf{K}}$ one obtains an expression of the type $[(\widehat{K}_{a,a} + \widehat{K}_{b,b})\mathbf{I} - \widehat{\mathbf{L}}]\widehat{\mathbf{y}}_{a,b}$, which is singular when $(\widehat{K}_{a,a} + \widehat{K}_{b,b})$ is an eigenvalue of $\widehat{\mathbf{L}}$. The normal form of the invariant manifold precisely resolves this singularity by requiring $\widehat{\mathbf{y}}_{a,b}$ to have a vanishing component along the singular direction. However, equation (45) still needs to be balanced along the singular directions, which can only be done by $\sum_{j=1}^{m} \mathbf{g}_{a,b}^{j}\widehat{\mathbf{v}}_j$. This defines the terms of the reduced representation $\mathbf{g}_{a,b}^{j}$ such that,

$$\mathbf{g}_{a,b}^{j} = \begin{cases} \widehat{\mathbf{w}}_j^{\dagger}(\widehat{\mathbf{f}}_{a,b} - \widehat{\mathbf{h}}_{a,b}); & \textit{if } \widehat{\lambda}_j = (\widehat{K}_{a,a} + \widehat{K}_{b,b}) \\ 0 & \textit{otherwise.} \end{cases} \tag{48}$$

2.1.3 Asymptotic terms of $\mathcal{O}(|x|^3)$

At third order (and higher orders) one also obtains contributions from the lower order terms.

$$\begin{aligned} &\sum_{(3)}^{[1,m]} \sum_{j=1}^{m} \Big[\widehat{K}_{i_1,j} x_{i_2} x_{i_3} x_j + \widehat{K}_{i_2,j} x_{i_1} x_{i_3} x_j + \\ &\widehat{K}_{i_3,j} x_{i_1} x_{i_2} x_j \Big] \widehat{\mathbf{y}}_{i_1,i_2,i_3} + \\ &\sum_{(2)}^{[1,m]} \sum_{(2)}^{[1,m]} \left[\mathbf{g}_{j_1,j_2}^{i_1} x_{i_2} x_{j_1} x_{j_2} + \mathbf{g}_{j_1,j_2}^{i_2} x_{i_1} x_{j_1} x_{j_2} \right] \widehat{\mathbf{y}}_{i_1,i_2} \\ &+ \sum_{i=1}^{m} \sum_{(3)}^{[1,m]} \mathbf{g}_{j_1,j_2,j_3}^{i} \widehat{\mathbf{v}}_j x_{j_1} x_{j_2,x_3} \\ &- \sum_{(3)}^{[1,m]} \widehat{\mathbf{L}}\widehat{\mathbf{y}}_{i_1,i_2,1_3} x_{i_1} x_{i_2} x_{i_3} = \sum_{(3)}^{[1,m]} \widehat{\mathbf{f}}_{i_1,i_2,i_3} x_{i_1} x_{i_2} x_{i_3}. \end{aligned} \tag{49}$$

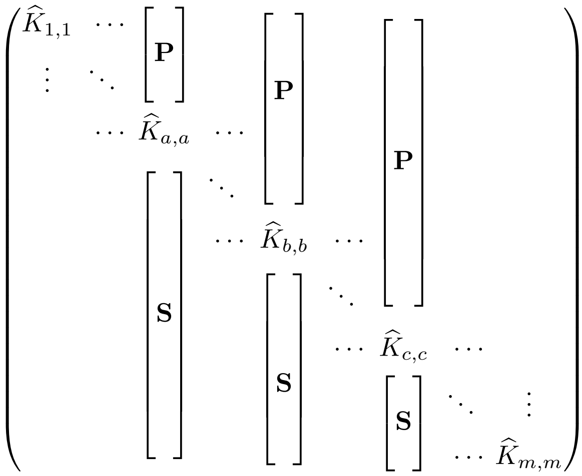

Fig. 2: Coupling of the vectors $\widehat{\mathbf{y}}_{i_1,i_2,i_3}$ with the $\widehat{\mathbf{K}}$ matrix for a given (a, b, c).

Collecting terms for a particular asymptotic term $x_a x_b$, where $(a, b) \in \mathcal{I}\{(2), [1, m]\}$, equation (49) reduces to,

$$\begin{aligned} &[\widehat{K}_{a,a} + \widehat{K}_{b,b}]\widehat{\mathbf{y}}_{a,b} + \Big[\sum_{\substack{i=1 \\ i\neq a}}^{b} \widehat{K}_{i,a}\widehat{\mathbf{y}}_{i,b} + \sum_{\substack{i=a \\ i\neq b}}^{m} \widehat{K}_{i,b}\widehat{\mathbf{y}}_{a,i} \Big] \\ &+ \sum_{j=1}^{m} \mathbf{g}_{a,b}^{j}\widehat{\mathbf{v}}_j - \widehat{\mathbf{L}}\widehat{\mathbf{y}}_{a,b} = \widehat{\mathbf{f}}_{a,b}. \end{aligned} \tag{50}$$

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